

The Picard group of $E^{hG_{24}}$

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Abstract

Let E be the height-two Morava E -theory associated to the supersingular elliptic curve over $\mathbf{F}_4$ at the prime 2. We compute the Picard group of $K(2)$ -local $E^{hG_{24}}$ -modules. Suspension generates a cyclic group of order 192. The proof obtains an Artin–Schreier d_{23} in the Picard descent spectral sequence from the C_4 calculation. Two finite Postnikov intervals determine the transfer needed for this differential. The Galois action on the remaining fifth filtration then excludes classes outside the suspension subgroup.

1 Introduction

Let E be the Morava E -theory of the formal group of the supersingular elliptic curve

$$C_0 : y^2 + y = x^3$$

over $\mathbf{F}_4$. Thus $\pi_0 E \cong W(\mathbf{F}_4)[[a]]$, where $W(\mathbf{F}_4)$ is the ring of 2-typical Witt vectors. The group $G_{24} = \text{Aut}_{\mathbf{F}_4}(C_0)$ acts on E . Write G_{48} for the extended automorphism group, with quotient $G_{48}/G_{24} = \text{Gal}(\mathbf{F}_4/\mathbf{F}_2)$. This quotient acts on $\text{Pic}(E^{hG_{24}})$ and fixes suspension classes. Throughout, $\text{Pic}(E^{hL})$, for $L \subseteq G_{24}$, denotes the Picard group of the category of $K(2)$ -local E^{hL} -modules. Galois descent identifies this with the Picard group of $K(2)$ -local L -equivariant E -modules; see [5, Sections 2–3]. The group law on Picard groups will be written additively. In particular, $n[\Sigma E^{hL}] = [\Sigma^n E^{hL}]$.

Theorem 1.1. *Suspension induces an isomorphism*

$$\mathbf{Z}/192 \xrightarrow{\sim} \text{Pic}(E^{hG_{24}}).$$

The exact suspension period of $E^{hG_{24}}$ is 192 [4, Remark 9.5]. It gives an injection $\mathbf{Z}/192 \hookrightarrow \text{Pic}(E^{hG_{24}})$. Surjectivity was asserted in [6, Proposition 13.23(2)]; the gap in the cited justification is recorded in [4, Remark 12.25 and footnote 1 on p. 74]. The proof below establishes surjectivity by descent.

The short Picard differentials leave possible additional classes in filtrations 5 and 7. The representation norm from C_4 to G_{24} identifies a nonzero transfer on the seventh source. We determine its target by a connecting map between finite Postnikov intervals. This supplies the transfer hypothesis of [5, Proposition 7.3] and gives a Picard d_{23} whose kernel is $\mathbf{F}_2$. Consequently the seventh filtration is generated by $[\Sigma^{96} E^{hG_{24}}]$. Finally, Witt Frobenius on the fifth filtration forces every remaining class to lie in the suspension subgroup.

2 Descent and preliminary calculations

For a subgroup $L \subseteq G_{24}$, write

$$\begin{aligned} B_2^{p,q}(L) &= H^p(L, E_q) \implies \pi_{q-p} E^{hL}, \\ P_2^{p,q}(L) &= H^p(L, \pi_q \text{pic}(E)) \implies \pi_{q-p} (\text{pic}(E)^{hL}) \end{aligned}$$

for the additive and Picard spectral sequences, where $E_q = \pi_q E$. Their differentials are denoted by d_r^+ and d_r^{pic} , respectively; both have bidegree $(r, r-1)$. For $q \geq 2$, the coefficient identification $\pi_q \text{pic}(E) = E_{q-1}$ identifies $P_2^{p,q}(L)$ with $B_2^{p,q-1}(L)$. For a cohomology class x in the latter group, write x^{pic} for its image in the former. When these representatives survive, their classes on page r are denoted by $[x]_r$ and $[x^{\text{pic}}]_r$. These brackets are defined only for surviving representatives; they do not assert a map from the entire second page to page r . In particular, $[\alpha x^{\text{pic}}]_r$ means that the coefficient is applied before passage to page r . The stable comparison and the first unstable differential are those of [10, Comparison Tool 5.2.4 and Theorem 6.1.1], restated in [5, Theorem 7.1].

For a spectrum V with L -action, V^{hL} denotes the homotopy fixed point spectrum, the limit of the action diagram $BL \rightarrow \text{Sp}$. Here Sp is the category of spectra. We use ordinary Postnikov truncations, taken equivariantly before homotopy fixed points. The connective cover $\tau_{\geq a} V$ has zero homotopy groups below degree a , and its canonical map to V induces isomorphisms in degrees at least a . The upper truncation $\tau_{\leq b} V$ has zero homotopy groups above degree b , and the canonical map $V \rightarrow \tau_{\leq b} V$ induces isomorphisms in degrees at most b . For integers $a \leq b$, the *Postnikov interval* is the spectrum

$$V[a, b] := \tau_{\geq a}(\tau_{\leq b} V), \quad \pi_q V[a, b] \cong \begin{cases} \pi_q V, & a \leq q \leq b, \\ 0, & q < a \text{ or } q > b. \end{cases} \quad (1)$$

Thus the endpoints a, b are inclusive homotopy degrees. For example, $E[16, 28]$ retains the homotopy groups of E in degrees 16 through 28, with the induced Postnikov data.

The descent filtration is

$$F^p \pi_m(V^{hL}) = \text{im}(\pi_m((\tau_{\geq m+p} V)^{hL}) \rightarrow \pi_m(V^{hL})).$$

All truncations precede homotopy fixed points. Taking $V = \text{pic}(E)$ and $L = G_{24}$ gives the descent filtration on $\text{Pic}(E^{hG_{24}})$. We write gr^p for the quotient of filtration p by filtration $p+1$. Write $B_r^{p,q}(L; [a, b])$ for the spectral sequence of $(E[a, b])^{hL}$, and $P_r^{p,q}(L; [a, b])$ for that of $(\text{pic}(E)[a, b])^{hL}$. In these spectral sequences we write $[x]_{r,[a,b]}$ and $[x^{\text{pic}}]_{r,[a,b]}$ for page classes. A class in a homotopy group is *detected* by a permanent page class when its image in the associated graded group is that class. We mark the homotopy lifts used below with a tilde.

Fix a subgroup $C_4 \subset G_{24}$, containing the central C_2 . For $K \subseteq L$ and an L -spectrum V , the stable transfer is

$$\text{tr}_K^L(V) : V^{hK} \rightarrow V^{hL}.$$

It is induced by the finite covering $BK \rightarrow BL$. We use $\text{tr}_K^L(V)_*$ for its map on homotopy groups and $\text{cor}_{K,r}^L$ for its map on spectral-sequence pages; the ambient spectral sequence will be specified. For $V = \text{pic}(E)$, the induced map on π_0 , under descent, is the equivariant norm

$$N_K^L : \text{Pic}(E^{hK}) \rightarrow \text{Pic}(E^{hL}).$$

If

$$J_L : \text{RO}(L) \rightarrow \text{Pic}(E^{hL}), \quad V \mapsto [E \wedge S^V]$$

is the representation-sphere homomorphism under descent, then

$$N_{C_4}^{G_{24}} J_{C_4}(V) = J_{G_{24}}(\text{Ind}_{C_4}^{G_{24}} V).$$

On the second page the transfer is ordinary cohomological corestriction. In the stable comparison range, the coefficient identification above intertwines additive and Picard page transfers [5, Sections 2–3 and 7.1].

2.1 Representation spheres and additive differentials

The representation calculation of [6, Propositions 12.11, 13.22, and 13.23], together with the exact suspension period, gives the factorization

$$\mathrm{RO}(G_{24}) \longrightarrow \mathbf{Z} \oplus \mathbf{Z}/8 \longrightarrow \frac{\mathbf{Z} \oplus \mathbf{Z}/8}{\langle (24, 1) \rangle} \cong \mathbf{Z}/192 \hookrightarrow \mathrm{Pic}(E^{hG_{24}}). \quad (2)$$

The first map is $V \mapsto (\dim V, \lambda(V))$, where $\lambda : \mathrm{RO}(G_{24}) \rightarrow \mathbf{Z}/8$ is additive. The displayed quotient describes the image of $J_{G_{24}}$; its final map is injective because suspension has exact order 192.

We use the G_{24} cohomology generators of [7, Theorem 2.15], with bidegrees

	η	ν	μ	$\bar{\kappa}$	c_4	c_6	Δ
(p, q)	$(1, 2)$	$(1, 4)$	$(1, 6)$	$(4, 24)$	$(0, 8)$	$(0, 12)$	$(0, 24)$

Here $j = c_4^3/\Delta$. The presentation for G_{24} is obtained from that for G_{48} by the Galois-equivariant extension of coefficients to $W(\mathbf{F}_4)$ [7, Lemma 1.32 and Remarks 1.39–1.40]. Set

$$z = \eta^3 \bar{\kappa} \Delta^{-1} \in B_2^{7,6}(G_{24}), \quad y = \eta^2 \bar{\kappa}^7 \Delta^{-6} \in B_2^{30,28}(G_{24}).$$

Their images support the nonzero additive differential

$$d_{23}^+([z]_{23}) = [y]_{23}. \quad (3)$$

In the Q_8 presentation of [8, Section 3.1], the classes D and g have bidegrees $(0, 8)$ and $(4, 24)$, with $\mathrm{res}(\Delta) = D^3$ and $\mathrm{res}(\bar{\kappa}) = g$. To verify (3), [8, Corollary 4.22(2)] gives

$$d_{23}^+([D^5 \eta^3]_{23}) = [D^{-10} g^6 \eta^2]_{23}.$$

The classes $D^{\pm 8}$ and g are permanent [8, Proposition 4.1 and Lemma 4.9]. In the Tate spectral sequence gD^{-8} is also invertible, so multiplication by it gives the nonzero differential

$$d_{23}^+([\eta^3 g D^{-3}]_{23}) = [\eta^2 g^7 D^{-18}]_{23}.$$

Both endpoints have positive filtration, 7 and 30. Hence [8, Lemma 2.6 and Method 2.7] give this differential in the homotopy fixed point spectral sequence. Since $Q_8 \triangleleft G_{24}$ has index 3, taking C_3 -invariants is exact. Under the comparison [8, Proposition 4.2 and Corollary 4.3], restriction sends Δ to D^3 and $\bar{\kappa}$ to g , giving (3).

For C_4 , we use the presentation of [5, Proposition 5.10]. Its generators $\eta, \nu, \varsigma, \varpi, \Delta_1$ have respective bidegrees

$$(1, 2), \quad (1, 4), \quad (1, 6), \quad (2, 8), \quad (0, 8).$$

We write $\mu_0 \in H^0(C_4, E_0)$ for the deformation parameter called μ in [5]. This distinguishes it from the class $\mu \in H^1(G_{24}, E_6)$ above. Put

$$z_{C_4} = \varsigma \varpi^3 \Delta_1^{-3} \in B_2^{7,6}(C_4), \quad w_{C_4} = \nu^2 \varpi^8 \Delta_1^{-7} \in B_2^{18,16}(C_4).$$

The differentials computed in [5, Section 7.2] are

$$\begin{aligned} d_{11}^+([\alpha z_{C_4}]_{11}) &= [\alpha w_{C_4}]_{11}, \\ d_{11}^{\mathrm{pic}}([\alpha z_{C_4}^{\mathrm{pic}}]_{11}) &= [(\alpha + \alpha^2) w_{C_4}^{\mathrm{pic}}]_{11}, \quad \alpha \in \mathbf{F}_4. \end{aligned} \quad (4)$$

The additive presentation applies to the $W(\mathbf{F}_4)$ -model by [8, Sections 2.3–2.4, Proposition 2.14]. Although [5] states its main Picard theorem over $\overline{\mathbf{F}}_2$, the two entries in (4) are copies of $\mathbf{F}_4$ on the E_{11} -page, and their maps under extension to $\overline{\mathbf{F}}_2$ are injective. Naturality therefore gives the displayed Picard formula over $\mathbf{F}_4$.

3 The low descent filtration

Let $\bar{b}$ denote reduction modulo 2, and define the additive map

$$\text{AS}_4 : W(\mathbf{F}_4)/4 \longrightarrow \mathbf{F}_4, \quad b \longmapsto \bar{b} + \bar{b}^2. \quad (5)$$

As an abelian group, $\ker(\text{AS}_4) \cong \mathbf{Z}/4 \oplus \mathbf{Z}/2$.

Proposition 3.1. *The descent filtration on $\text{Pic}(E^{hG_{24}})$ satisfies*

$$\begin{aligned} \text{Pic}(E^{hG_{24}})/F^5 \text{Pic}(E^{hG_{24}}) &\cong \mathbf{Z}/24, \\ F^6 \text{Pic}(E^{hG_{24}}) &= F^7 \text{Pic}(E^{hG_{24}}), \quad F^8 \text{Pic}(E^{hG_{24}}) = 0. \end{aligned} \quad (6)$$

The quotient $\text{Pic}(E^{hG_{24}})/F^5 \text{Pic}(E^{hG_{24}})$ is generated by the image of $[\Sigma E^{hG_{24}}]$. Moreover, $\text{gr}^5 \text{Pic}(E^{hG_{24}})$ embeds in $\ker(\text{AS}_4)$, and $\text{gr}^7 \text{Pic}(E^{hG_{24}})$ is a subquotient of $\mathbf{F}_4$. In particular, $[\Sigma^{96} E^{hG_{24}}]$ is a nonzero element of $F^7 \text{Pic}(E^{hG_{24}})$.

Proof. The algebraic line-bundle term is $H^1(G_{24}, E_0^\times)$. On the standard deformation chart [2, Section 3.1],

$$\mathcal{C} : y^2 + axy + y = x^3, \quad j = \frac{a^3(a^3 - 24)^3}{a^3 - 27}.$$

Consider the invariant subring $W(\mathbf{F}_4)[[j]] \subseteq E_0$. The monic equation

$$a^3(a^3 - 24)^3 - j(a^3 - 27) = 0$$

reduces to a^{12} modulo $(2, j)$, so E_0 is finite free of rank 12 over $W(\mathbf{F}_4)[[j]]$. The central C_2 acts trivially on E_0 , and the action of $G_{24}/C_2 \cong A_4$ is faithful. The degree of the extension of fraction fields is 12, so $\text{Frac}(E_0)^{A_4} = \text{Frac}(W(\mathbf{F}_4)[[j]])$. Normality of $W(\mathbf{F}_4)[[j]]$ then gives $E_0^{A_4} = W(\mathbf{F}_4)[[j]]$.

Both rings are factorial. Hilbert 90 and the divisor sequence yield

$$H^1(A_4, E_0^\times) \cong \frac{\text{Div}(E_0)^{A_4}}{\text{div}(\text{Frac}(W(\mathbf{F}_4)[[j]])^\times)} \cong \bigoplus_{\mathfrak{q} \in (\text{Spec } W(\mathbf{F}_4)[[j]])^{(1)}} \mathbf{Z}/e_{\mathfrak{q}},$$

where $e_{\mathfrak{q}}$ is the ramification index above $\mathfrak{q}$. The two nontrivial indices are 3 over $j = 0$ and 2 over $j = 1728$. Indeed,

$$\begin{aligned} \frac{dj}{da} &= \frac{9a^2(a^3 - 24)^2(a^6 - 36a^3 + 216)}{(a^3 - 27)^2}, \\ j - 1728 &= \frac{(a^6 - 36a^3 + 216)^2}{a^3 - 27}. \end{aligned}$$

There is no other horizontal branch divisor, and the vertical prime (2) has ramification index 1. Thus $H^1(A_4, E_0^\times) \cong \mathbf{Z}/6$. Inflation–restriction for the central C_2 bounds $H^1(G_{24}, E_0^\times)$ by a group of order 12. The Hodge line has order divisible by 6 and 4, from its characters on the characteristic-zero geometric inertia groups C_6 and C_4 ; the discriminant trivializes its twelfth power. It therefore generates

$$H^1(G_{24}, E_0^\times) \cong \mathbf{Z}/12.$$

The Hodge line and the parity class are associated-graded images of suspension classes; together they give a cyclic quotient of order 24.

The short Picard differentials are obtained from [10, Section 8.2] and [1, Sections 6.4–6.6] by applying their formulas to the completed coefficient ring. In filtration 3 the differential has the form

$$\mathbf{F}_4[[j]] \longrightarrow \mathbf{F}_4[[j]], \quad f \longmapsto f + jf^2.$$

Its kernel is zero, since $1 + jf$ is a unit. In filtration 5, the coefficient generator is

$$\nu\bar{\kappa}\Delta^{-1} \in H^5(G_{24}, E_4).$$

The cohomology presentation gives $H^5(G_{24}, E_4) = (W(\mathbf{F}_4)/4)\{\nu\bar{\kappa}\Delta^{-1}\}$, and the first unstable differential has coefficient $\bar{b} + \bar{b}^2$. The kernel on coefficients is $\ker(\text{AS}_4)$. No later differential enters this position: the possible d_3 source $H^2(G_{24}, E_2)$ is zero, the possible d_5 source consists of permanent invariant units, and the other sources vanish. Thus passage to the coefficient of $(\nu\bar{\kappa}\Delta^{-1})^{\text{pic}}$ defines an injection

$$\iota_5 : \text{gr}^5 \text{Pic}(E^{hG_{24}}) \hookrightarrow \ker(\text{AS}_4).$$

The only other possible positive diagonal filtration is 7, with coefficient group a subquotient of $\mathbf{F}_4$; see [1, Proposition 6.8 and Sections 6.5–6.7].

We also need separatedness of the filtration. The proof of [1, Proposition 6.8], applied to the completed additive calculation, gives

$$P_{24}^{p,p}(G_{24}) = 0 \quad (p > 7).$$

Each such entry is killed by a stable-range differential of length at most 23. In addition,

$$P_{24}^{p,p+1}(G_{24}) = 0 \quad (p > 0).$$

For the second assertion, the nonnegative-total-degree comparison [5, Theorem 7.1(1)] identifies both incoming and outgoing differentials with those in additive total degree zero. The additive column has no positive-filtration survivors, and all its differentials have length at most 23 [1, Proposition 6.6].

Here the uniform finite page is essential. For the lower-tail tower

$$T_n = (\tau_{\geq n} \text{pic}(E))^{hG_{24}}, \quad \text{holim}_n T_n = 0,$$

the two adjacent vanishings make the degree-zero stable images in the 24th derived exact couple eventually constant. This image tower is obtained from $\pi_0 T_n$ by a bounded shift and passage to images, so it has the same inverse limit. The Milnor exact sequence and $\text{holim}_n T_n = 0$ give that limit as zero. The image tower is therefore zero, making the degree-zero tail pro-zero. Consequently the filtration on $\text{Pic}(E^{hG_{24}})$ is separated and eventually zero. The vanishing of every $\text{gr}^p \text{Pic}(E^{hG_{24}})$ for $p > 7$ now gives $F^8 \text{Pic}(E^{hG_{24}}) = 0$.

It follows that $[\Sigma^{24} E^{hG_{24}}] \in F^5 \text{Pic}(E^{hG_{24}})$. Since $\ker(\text{AS}_4)$ has exponent 4, $[\Sigma^{96} E^{hG_{24}}] \in F^7 \text{Pic}(E^{hG_{24}})$; its nonzero value follows from the exact order 192 of $[\Sigma E^{hG_{24}}]$. $\square$

4 The Picard differential

The remaining seventh-filtration calculation is the following.

Proposition 4.1. *In the Picard spectral sequence for G_{24} ,*

$$d_{23}^{\text{pic}}([\alpha z^{\text{pic}}]_{23}) = [(\alpha + \alpha^2)y^{\text{pic}}]_{23}, \quad \alpha \in \mathbf{F}_4. \quad (7)$$

To prove the proposition, we determine the transfer of the actual target of the C_4 differential (4). We use three Postnikov intervals in the sense of (1):

$$\begin{aligned} E[16, 28] &= \tau_{\geq 16} \tau_{\leq 28} E, \\ E[4, 28] &= \tau_{\geq 4} \tau_{\leq 28} E, \\ E[4, 15] &= \tau_{\geq 4} \tau_{\leq 15} E. \end{aligned}$$

They retain degrees 16–28, 4–28, and 4–15, respectively. The Postnikov projection from the middle interval to the lower one has the upper interval as its fiber. This gives the fiber sequence of spectra with L -action

$$E[16, 28] \longrightarrow E[4, 28] \longrightarrow E[4, 15] \longrightarrow \Sigma E[16, 28].$$

The first arrow is the connective-cover map; the second is the Postnikov projection; and the last is the boundary map, with Σ denoting suspension. For $L = C_4, G_{24}$, homotopy fixed points preserve this fiber sequence. Its long exact sequence of homotopy groups contains the connecting homomorphism

$$\delta_L : \pi_{-1}((E[4, 15])^{hL}) \longrightarrow \pi_{-2}((E[16, 28])^{hL}). \quad (8)$$

For a class in the source, the vanishing of δ_L is equivalent to the existence of a lift to $\pi_{-1}((E[4, 28])^{hL})$. These connecting maps commute with the transfers on homotopy groups.

Lemma 4.2. *The edge homomorphisms give isomorphisms*

$$\begin{aligned} F^7 \pi_{-1}((E[4, 15])^{hG_{24}}) &\xrightarrow{\sim} \mathbf{F}_4\{[z]_{\infty, [4, 15]}\}, \\ \pi_{-2}((E[16, 28])^{hC_4}) &\xrightarrow{\sim} \mathbf{F}_4\{[w_{C_4}]_{\infty, [16, 28]}\}. \end{aligned}$$

In particular,

$$F^8 \pi_{-1}((E[4, 15])^{hG_{24}}) = 0, \quad F^{19} \pi_{-2}((E[16, 28])^{hC_4}) = 0.$$

Let $\tilde{z}$ and $\tilde{w}_{C_4}$ be the respective inverse images of the displayed generators. These are homotopy classes, with scalar structures transported through the edge isomorphisms. Choose a homotopy class $\tilde{z}_{C_4} \in F^7 \pi_{-1}((E[4, 15])^{hC_4})$ detected by $[z_{C_4}]_{\infty, [4, 15]}$. Then $\tilde{w}_{C_4} = \delta_{C_4}(\tilde{z}_{C_4})$.

Proof. In $B_2^{p,q}(G_{24}; [4, 15])$, the possible entries in total degree -1 and filtration at least 7 are

(p, q)	representative	effect of d_3
(7, 6)	$\eta^3 \bar{\kappa} \Delta^{-1}$	quotient by j
(9, 8), (13, 12)	0	
(11, 10)	$\mu \eta^2 \bar{\kappa}^2 \Delta^{-2}$	injective outgoing map
(15, 14)	$\eta^7 \bar{\kappa}^2 \Delta^{-2}$	incoming boundary.

The nonzero entries have coefficients in $\mathbf{F}_4[[j]]$ before taking the indicated quotient. The relevant formulas are

$$\begin{aligned} d_3^+([c_4^2 c_6 \bar{\kappa} \Delta^{-2}]_{3, [4, 15]}) &= [j \eta^3 \bar{\kappa} \Delta^{-1}]_{3, [4, 15]}, \\ d_3^+([\mu \eta^2 \bar{\kappa}^2 \Delta^{-2}]_{3, [4, 15]}) &= [\eta^6 \bar{\kappa}^2 \Delta^{-2}]_{3, [4, 15]}, \\ d_3^+([\mu \eta^3 \bar{\kappa}^2 \Delta^{-2}]_{3, [4, 15]}) &= [\eta^7 \bar{\kappa}^2 \Delta^{-2}]_{3, [4, 15]}. \end{aligned}$$

Their source and target bidegrees are respectively

$$(4, 4) \rightarrow (7, 6), \quad (11, 10) \rightarrow (14, 12), \quad (12, 12) \rightarrow (15, 14);$$

all are retained in the interval. These identities follow from the cohomology relation $c_4 \bar{\kappa} = \Delta \eta^4$ and the full additive differentials $d_3^+([c_6]_3) = [c_4 \eta^3]_3$ and $d_3^+([\mu]_3) = [\eta^4]_3$ [7, Theorem 2.15 and Lemma 2.21]. The surviving copy of $\mathbf{F}_4$ at (7, 6) has no outgoing differential in this interval: its d_5 and d_9 targets are zero, and its d_7 target at (14, 12) is already a d_3 -boundary. The incoming maps removed by the lower cutoff have zero source $H^2(G_{24}, E_2)$, or arise from permanent classes in degree zero. This proves the assertion about $(E[4, 15])^{hG_{24}}$.

In $B_2^{p,q}(C_4; [16, 28])$, the only possible entries in total degree -2 are

(p, q)	representative	survival
(18, 16)	$\varpi^9 \Delta_1^{-7}$	d_3 leaves its double
(22, 20)	$\eta^2 \varpi^{10} \Delta_1^{-8}$	d_3 boundary
(26, 24)	$\varpi^{13} \Delta_1^{-10}$	d_3 leaves its double; d_5 kills it
(30, 28)	$\eta^2 \varpi^{14} \Delta_1^{-11}$	d_3 boundary.

There are no entries at filtrations 20, 24, 28, and odd filtrations have odd internal degree. The formulas of [5, Propositions 5.10, 5.21, and 5.24] give

$$\begin{aligned}
d_3^+([\varpi^9 \Delta_1^{-7}]_{3,[16,28]}) &= [\eta \varpi^{10} \Delta_1^{-8}]_{3,[16,28]}, \\
d_3^+([\eta \varpi^9 \Delta_1^{-7}]_{3,[16,28]}) &= [\eta^2 \varpi^{10} \Delta_1^{-8}]_{3,[16,28]}, \\
d_3^+([\varpi^{13} \Delta_1^{-10}]_{3,[16,28]}) &= [\eta \varpi^{14} \Delta_1^{-11}]_{3,[16,28]}, \\
d_3^+([\eta \varpi^{13} \Delta_1^{-10}]_{3,[16,28]}) &= [\eta^2 \varpi^{14} \Delta_1^{-11}]_{3,[16,28]}, \\
d_5^+([\nu \varpi^{10} \Delta_1^{-8}]_{5,[16,28]}) &= [2 \varpi^{13} \Delta_1^{-10}]_{5,[16,28]}.
\end{aligned}$$

The outgoing d_3 targets are (21, 18) and (29, 26); the incoming d_3 sources are (19, 18) and (27, 26). All lie in the interval. The coefficient module of either odd power of ϖ is

$$W(\mathbf{F}_4)[[\mu_0]]/(4, 2\mu_0).$$

Its d_3 -kernel consists of the $\mathbf{F}_4$ -multiples of the double. Here $\nu^2 = 2\varpi$ and $\mu_0\nu = 0$ imply $2\mu_0\varpi = 0$.

The d_5 source has bidegree (21, 20). Its possible incoming d_3 source at (18, 18) is zero, and its outgoing d_3 is zero because $\eta\nu = 0$. It therefore exists on the E_5 -page. The displayed d_5 follows from the permanence of $\varpi^2 \Delta_1$, the formula $d_5^+([\Delta_1]_5) = [\nu \Delta_1^{-1} \varpi^2]_5$, and the cohomology relation $\nu^2 = 2\varpi$. Thus only the following filtration-18 copy can remain:

$$\mathbf{F}_4\{[2\varpi^9 \Delta_1^{-7}]_{6,[16,28]}\} = \mathbf{F}_4\{[w_{C_4}]_{6,[16,28]}\}.$$

The representative z_{C_4} survives to $[z_{C_4}]_{\infty,[4,15]}$, since its first outgoing differential is the d_{11} in (4), with target above this interval. The lower cutoff 4 deletes only the possible incoming d_5 from $H^2(C_4, E_2) = 0$ and the d_7 from permanent degree-zero classes. At the target (18, 16), an incoming map from internal degree below 4 has length at least 15. Consequently the same d_{11} occurs for $E[4, 28]$. The Postnikov description of differentials [5, Lemma 7.2] now shows that $\delta_{C_4}(\tilde{z}_{C_4})$ is detected by $[w_{C_4}]_{\infty,[16,28]}$. It is nonzero, proving the assertion about $(E[16, 28])^{hC_4}$. Each interval is finite, so its filtration in a fixed total degree is finite. $\square$

Lemma 4.3. *On the E_4 -page of the additive spectral sequence, transfer is*

$$\text{cor}_{C_4,4}^{G_{24}} : B_4^{7,6}(C_4) \longrightarrow B_4^{7,6}(G_{24}), \quad [z_{C_4}]_4 \longmapsto [z]_4.$$

Proof. The C_2 Picard calculation over $\mathbf{F}_4$ places $[\Sigma^8 E^{hC_2}]$ in filtration 7: parity and the algebraic line have total order 4, the third filtration has exponent 2, and the fifth term is zero. These are the height-two case of [9, Proposition 3.9, Lemma 3.11, Theorem 3.12, and Figure 2]; the finite-field model and its extension to the elliptic formal group are specified in [9, Section 2.1]. Let σ_{C_4} denote the sign representation of C_4 . The norm of this suspension is

$$N_{C_2}^{C_4}([\Sigma^8 E^{hC_2}]) = J_{C_4}(8\mathbf{1} + 8\sigma_{C_4}) = [\Sigma^{16} E^{hC_4}],$$

where the last equality uses the $4 - 4\sigma_{C_4}$ periodicity of [8, Lemma 2.21]. The norm preserves filtration. Exact period 32 and vanishing above filtration 7 show that $[\Sigma^{16} E^{hC_4}]$ is a

nonzero seventh-filtration class. Formula (4) makes its leading coefficient the nonzero element of $\ker(\alpha \mapsto \alpha + \alpha^2) = \mathbf{F}_2$. It is therefore detected by $[z_{C_4}^{\text{pic}}]_\infty$.

Let $\rho = \text{Ind}_{C_4}^{G_{24}} \mathbf{1}$. It has dimension 6, and (2) gives

$$N_{C_4}^{G_{24}}([\Sigma^{16} E^{hC_4}]) = J_{G_{24}}(16\rho) = [\Sigma^{96} E^{hG_{24}}],$$

since $16\lambda(\rho) = 0$ in $\mathbf{Z}/8$. Proposition 3.1 makes its seventh-filtration leading term nonzero.

We identify this leading term on the E_4 -page. For G_{24} , Lemma 4.2 gives $B_4^{7,6}(G_{24}) = \mathbf{F}_4\{[z]_4\}$. For C_4 , the relation $\eta^3 = \mu_0 \varsigma \varpi \Delta_1^{-1}$ gives

$$d_3^+([\eta^2 \varpi \Delta_1^{-1}]_3) = [\eta^3 \varpi^2 \Delta_1^{-2}]_3 = [\mu_0 z_{C_4}]_3.$$

It follows that $B_4^{7,6}(C_4) = \mathbf{F}_4\{[z_{C_4}]_4\}$. The relevant d_3 's are in the stable comparison range, so the coefficient comparison gives

$$B_4^{7,6}(L) \cong P_4^{7,7}(L), \quad L = C_4, G_{24},$$

and intertwines their $\mathbf{F}_4$ -linear page transfers. The nonzero norm computed above forces $\text{cor}_{C_4,4}^{G_{24}}$ to be nonzero.

To determine its scalar, note that after E_4 no differential enters $P^{7,7}(G_{24})$. The only possible odd lengths are 5 and 7: their sources are $H^2(G_{24}, E_2) = 0$ and permanent invariant units, respectively. Even lengths have zero coefficient groups, and lengths greater than 7 have negative source filtration. Thus $P_\infty^{7,7}(G_{24})$ embeds in $P_4^{7,7}(G_{24})$. Write the transfer on this page as

$$\text{cor}_{C_4,4}^{G_{24}}([z_{C_4}]_4) = c[z]_4, \quad c \in \mathbf{F}_4^\times.$$

The monomial z comes from the G_{48} presentation and is fixed by Witt Frobenius. The suspension $[\Sigma^{96} E^{hG_{24}}]$ is also fixed. Hence $c^2 = c$, and therefore $c = 1$. $\square$

Lemma 4.4. *There is a nonzero class $\tilde{y} \in F^{30}\pi_{-2}((E[16, 28])^{hG_{24}})$, detected by $[y]_{\infty, [16, 28]}$, such that*

$$\text{tr}_{C_4}^{G_{24}}(E[16, 28])_*(\tilde{w}_{C_4}) = \tilde{y}.$$

Proof. By (3), the representative z survives through Postnikov degree 27, and the obstruction in degree 28 is represented by $[y]_{23}$. The same assertion holds after the lower cutoff 4. Indeed, the target of d_r from (7, 6) has internal degree $5 + r$; an incoming map to that target from internal degree below 4 has length at least $r + 3$. At the source, the d_3 that removes the j -multiples is retained. The removed d_5 source is zero, and the removed d_7 source consists of permanent degree-zero classes. The comparisons therefore preserve the obstruction through d_{23} .

Choose a representative in $\pi_{-1}((E[4, 27])^{hG_{24}})$. Its image in $\pi_{-1}((E[4, 15])^{hG_{24}})$ is the unique class $\tilde{z}$ of Lemma 4.2 detected by $[z]_{\infty, [4, 15]}$. Since it already lifts through degree 27,

$$\delta_{G_{24}}(\tilde{z}) \in F^{30}\pi_{-2}((E[16, 28])^{hG_{24}}).$$

This boundary is nonzero. A zero boundary would lift $\tilde{z}$ to $\pi_{-1}((E[4, 28])^{hG_{24}})$. Its projection to $\pi_{-1}((E[4, 5])^{hG_{24}})$ would still be zero, so the lift would remain in F^7 . Mapping it to $\pi_{-1}((\tau_{\leq 28} E)^{hG_{24}})$ would then lift $[z]_{23}$ across the nonzero differential (3), a contradiction.

Detection in bidegree (30, 28) is compared along the map

$$(E[16, 28])^{hG_{24}} \longrightarrow (E[4, 28])^{hG_{24}}.$$

Up to E_{14} the target comparison is unchanged. Before E_{23} , the remaining potential incoming maps from the deleted lower interval are

r	15	17	19	21
source	(15, 14)	(13, 12)	(11, 10)	(9, 8).

All four source groups have vanished by E_4 , as computed in Lemma 4.2. The target comparison is therefore an isomorphism on page 23. In the upper interval the representative y is permanent, and the boundary is detected by $[y]_{\infty, [16, 28]}$. In the full additive spectral sequence, by contrast, $[y]_{23}$ is the boundary in (3). There is no filtration above 30 in this degree of $(E[16, 28])^{hG_{24}}$. Put $\tilde{y} = \delta_{G_{24}}(\tilde{z})$.

Lemma 4.3 identifies the transferred leading term of $\tilde{z}_{C_4}$ with $[z]_{\infty, [4, 15]}$. There is no higher-filtration ambiguity by Lemma 4.2, so

$$\mathrm{tr}_{C_4}^{G_{24}}(E[4, 15])_*(\tilde{z}_{C_4}) = \tilde{z}.$$

Naturality of (8) now gives

$$\begin{aligned} \mathrm{tr}_{C_4}^{G_{24}}(E[16, 28])_*(\tilde{w}_{C_4}) &= \mathrm{tr}_{C_4}^{G_{24}}(E[16, 28])_*\delta_{C_4}(\tilde{z}_{C_4}) \\ &= \delta_{G_{24}}(\mathrm{tr}_{C_4}^{G_{24}}(E[4, 15])_*(\tilde{z}_{C_4})) \\ &= \delta_{G_{24}}(\tilde{z}) = \tilde{y} \end{aligned}$$

in actual homotopy groups. $\square$

Proof of Proposition 4.1. Choose $\omega \in \mathbf{F}_4$ with $\omega^2 + \omega = 1$. The inequality $28 \leq 2 \cdot 16 - 1$ gives a natural equivalence

$$\mathrm{pic}(E)[17, 29] \simeq \Sigma \mathrm{gl}_1(E)[16, 28] \simeq \Sigma E[16, 28]$$

compatible with transfer [10, Corollary 5.2.3]. Consequently, for $L = C_4, G_{24}$, it induces an isomorphism

$$\pi_{-1}((\mathrm{pic}(E)[17, 29])^{hL}) \cong \pi_{-2}((E[16, 28])^{hL}).$$

The C_4 differential $d_{11}^{\mathrm{pic}}([\omega z_{C_4}^{\mathrm{pic}}]_{11}) = [w_{C_4}^{\mathrm{pic}}]_{11}$ has an actual target in the group on the left. Under the displayed isomorphism this target corresponds to $\tilde{w}_{C_4}$: Lemma 4.2 leaves no higher-filtration ambiguity.

Apply [5, Proposition 7.3] with parameters

$$(a, b, c) = (4, 16, 28), \quad (s, m, p) = (18, -2, 12), \quad r = 6, \quad r' = 11.$$

Here $r' \geq r$, $r' > m + s + 1 - b = 1$, and $c \leq 2b - 1$. Lemma 4.4 supplies the required equality of actual transfers in the upper interval. The resulting differential in $\mathrm{pic}(E)[5, 29]$ is

$$d_{23}^{\mathrm{pic}, [5, 29]}([\omega z^{\mathrm{pic}}]_{23, [5, 29]}) = [y^{\mathrm{pic}}]_{23, [5, 29]}.$$

Compare the truncated and full Picard spectral sequences via

$$\mathrm{pic}(E)[5, 29] \longrightarrow \tau_{\leq 29} \mathrm{pic}(E) \longleftarrow \mathrm{pic}(E).$$

At the source (7, 7), the incoming d_3 is retained; the removed d_5 source is $H^2(G_{24}, E_2) = 0$, and the removed d_7 source consists of permanent invariant units. At (30, 29), an incoming map from internal degree below 5 has length at least 26. The upper cutoff cannot create an earlier incoming boundary there, since all earlier outgoing maps from a candidate incoming source have internal degree below 29. The relevant outgoing differentials of the target are in the stable comparison range and vanish, as in the additive spectral sequence. The displayed d_{23} therefore holds in the full Picard spectral sequence.

The suspension class $[\Sigma^{96} E^{hG_{24}}]$ is detected by $[z^{\mathrm{pic}}]_{\infty}$, so $d_{23}^{\mathrm{pic}}([z^{\mathrm{pic}}]_{23}) = 0$. The differentials are $\mathbf{F}_2$ -linear; the two coefficients $1, \omega$ now give (7) for all $\alpha \in \mathbf{F}_4$. $\square$

5 The Galois action on the fifth filtration

The extended group G_{48} induces an action of $\text{Gal}(\mathbf{F}_4/\mathbf{F}_2)$ on $\text{Pic}(E^{hG_{24}})$. Let σ be its nontrivial element. Then $\sigma^2 = 1$, and σ fixes every suspension. The class $\nu\bar{\kappa}\Delta^{-1}$ used in Proposition 3.1 comes from G_{48} . The coefficient identification

$$H^5(G_{24}, E_4) = (W(\mathbf{F}_4)/4)\{\nu\bar{\kappa}\Delta^{-1}\}$$

is therefore Galois equivariant, with σ acting on the coefficients by Witt Frobenius [7, Lemma 1.32, Remarks 1.39–1.40, and Theorem 2.15]. In particular, the injection ι_5 constructed in Proposition 3.1 is equivariant. Denote its composite with the quotient homomorphism by

$$\ell_5 : F^5 \text{Pic}(E^{hG_{24}}) \longrightarrow \ker(\text{AS}_4).$$

Thus ℓ_5 records the coefficient of the fifth-filtration leading term, and

$$\ker(\ell_5) = F^6 \text{Pic}(E^{hG_{24}}) = F^7 \text{Pic}(E^{hG_{24}}).$$

Proof of Theorem 1.1. Proposition 4.1 bounds the seventh filtration by

$$\ker(\alpha \mapsto \alpha + \alpha^2) = \mathbf{F}_2.$$

Proposition 3.1 supplies the nonzero class $[\Sigma^{96} E^{hG_{24}}]$, so

$$F^7 \text{Pic}(E^{hG_{24}}) = \langle [\Sigma^{96} E^{hG_{24}}] \rangle \cong \mathbf{Z}/2.$$

The subgroup $\langle [\Sigma^{24} E^{hG_{24}}] \rangle$ has order 8. Its image under ℓ_5 has order 4 and is fixed by σ . Since

$$\ker(\text{AS}_4)^\sigma = W(\mathbf{F}_2)/4 = \mathbf{Z}/4,$$

the image is exactly this constant subgroup. In particular,

$$\ell_5([\Sigma^{48} E^{hG_{24}}]) = 2 \in W(\mathbf{F}_4)/4.$$

The quotient $\ker(\text{AS}_4)/(\mathbf{Z}/4)$ has order 2. If an additional fifth-filtration class survived, choose $x \in F^5 \text{Pic}(E^{hG_{24}})$ for which $\ell_5(x)$ represents its nonzero element. Modulo constants, $\ell_5(x) = 2[\omega]$, where $\omega^2 + \omega = 1$ and $[\omega]$ denotes the Teichmüller lift. Since $2([\omega^2] - [\omega]) = 2$ modulo 4,

$$(\sigma - 1)\ell_5(x) = 2 \quad \text{in the constant subgroup } \mathbf{Z}/4.$$

Equivariance and the description of $\ker(\ell_5)$ imply

$$(\sigma - 1)x \in [\Sigma^{48} E^{hG_{24}}] + F^7 \text{Pic}(E^{hG_{24}}).$$

Therefore

$$(\sigma - 1)x = [\Sigma^{48} E^{hG_{24}}] \quad \text{or} \quad [\Sigma^{144} E^{hG_{24}}].$$

Both right sides are fixed by σ . Applying $\sigma + 1$ gives

$$0 = (\sigma^2 - 1)x = (\sigma + 1)(\sigma - 1)x = [\Sigma^{96} E^{hG_{24}}],$$

contrary to the exact order of $[\Sigma E^{hG_{24}}]$. Thus $F^5 \text{Pic}(E^{hG_{24}}) = \langle [\Sigma^{24} E^{hG_{24}}] \rangle$. The quotient by this filtration subgroup is generated by the image of $[\Sigma E^{hG_{24}}]$, and the filtration subgroup itself consists of suspension classes. Hence

$$\text{Pic}(E^{hG_{24}}) = \langle [\Sigma E^{hG_{24}}] \rangle \cong \mathbf{Z}/192. \quad \square$$

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